1. An amount of $4500 is invested at 14% interest, compounded daily.

   a. Find the future value in 7 years.
   b. Determine whether more or less money would have accrued if the money had been compounded quarterly.

   a. The total amount is $[ ].
   (Round to the nearest cent as needed.)

   b. Determine whether more or less money would have accrued if the money had been compounded quarterly. Choose the correct answer below.

   ○ more
   ○ less

2. If $12,000 is invested at 12% interest compounded quarterly, find the interest earned in 10 years.

   The interest earned in 10 years is $[ ].
   (Do not round until the final answer. Then round to two decimal places as needed.)

3. If $3000 is invested at 3% compounded continuously, what is the amount after 9 years?

   The amount after 9 years will be $[ ].
   (Round to the nearest cent.)

4. What lump sum must be invested at 6%, compounded monthly, for the investment to grow to $59,000 in 10 years?

   The lump sum $[ ] invested at 6%, compounded monthly, grows to $59,000 in 10 years.
   (Do not round until the final answer. Then round to the nearest cent as needed.)

*5. Following the birth of a child, a parent wants to make an initial investment P_0 that will grow to $60,000 for the child's education at age 17. Interest is compounded continuously at 7%.
   What should the initial investment be? Such an amount is called the present value of $60,000 due 17 years from now.

   The present value is about $[ ].
   (Do not round until the final answer. Then round to two decimal places as needed.)

6. At the end of t years, the future value of an investment of $11,000 in an account that pays 6% APR compounded monthly is $S = 11{,}000\left(1 + \frac{0.06}{12}\right)^{12t}$ dollars. Assuming no withdrawals or additional deposits, how long will it take for the investment to amount to $44,000?

   The investment will grow to $44,000 in [ ] years.
   (Do not round until the final answer. Then round to two decimal places as needed.)

7. What rate of interest compounded annually is required to triple an investment in 26 years?

   The rate of interest required is [ ]%.
   (Round to two decimal places as needed.)

8. Suppose that $14,000 is invested at continuous rate r and grows to $81,104 in 28 years. Find the interest rate r.

   The interest rate is [ ]%.
   (Round to the nearest percent as needed.)